

The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). At the bottom, a taskbar is visible with several application icons, including what looks like a web browser, a file explorer, and some development tools. The overall image is out of focus, serving as a backdrop for the text.

**C#**

**Level 1**

# **Method Overloading**



C#

Level 1

# Method Overloading

# Definition

## Method Overloading

 Multiple methods can have the same name as long as their parameter lists are different.

 C# selects the appropriate overload based on the arguments passed.

# Explanation

## Different Number of Parameters

 Methods can differ by the number of parameters.

## Different Parameter Types

 Methods can also differ by the types of their parameters.

# Procedure

1. Define multiple methods with the same name.
2. Give each method a different parameter list.
3. Call the method using the required arguments.
4. C# selects the matching overload.

## 💡 Example

```
static int Sum(int Num1, int Num2)
{
    return Num1 + Num2;
}
static int Sum(int Num1, int Num2, int Num3)
{
    return Num1 + Num2 + Num3;
}
static int Sum(int Num1, int Num2, int Num3, int Num4)
{
    return Num1 + Num2 + Num3 + Num4;
}
```

### ➡ Result

Enter first number: 3  
Enter second number: 5  
Enter third number: 7  
Enter fourth number: 9  
Sum of 3 and 5 = 8  
Sum of 3, 5, and 7 = 15  
Sum of 3, 5, 7, and 9 = 24

## Note

- ◆ The parameter list includes the number, types, and order of parameters.
- ◆ The method name alone is not enough to determine which overload is used.

## ⊘ Warning

- ◆ Changing only the return type does not create an overload.
- ◆ These methods have identical parameter lists and cannot exist together.

# ✦✦ Summary

- ✓ Method overloading allows multiple methods to share the same name.
- ✓ Overloads must have different parameter lists.
- ✓ They can differ by parameter count or parameter types.
- ✓ C# selects the appropriate overload based on the arguments.

Thank you 😊